

Orlando, Florida, USA

Research Interests

Education

Computer Science, May 2021

Computer Science, May 2012

Professional Experience

Tasks: Develop systems for tutoring military personnel in a variety of day-to-day tasks; Work with a team of experts to create ecologically valid scenarios for trainees; Utilized Agile software development methodologies to manage deliverables within a cross-functional team

Research Experience

Graduate Research Assistant

University of Central Florida (Orlando, FL, USA)

Interactive Systems and User Experience Lab, December 2012 – May 2021

Director: Dr. Joseph LaViola

Tasks: Conduct gesture recognition research; Perform research paper writing; Oversee undergraduate researchers; Attend events to promote research lab and present work; Demonstrate research projects to lab visitors and potential sponsors

Research Intern

Microsoft Research (Redmond, WA, USA)

Multimedia, Interaction, & eXperiences Group, June 2015 – September 2015

Director: Zhengyou Zhang (Mentored by Nikolai Smolyanskiy)

Tasks: Conduct research with teleoperated drones; Simplify control schemes for consumer flying drones; Present research ideas at weekly meetings

Undergraduate Research Assistant

University of Central Florida (Orlando, FL, USA)

Center for Research in Computer Vision, May 2011 – October 2011

Director: Dr. Mubarak Shah

Tasks: Conduct computer vision research; Present research ideas and progress at weekly meetings; Assist in grant writing

Teaching

Instructor of Record

University of Central Florida (Orlando, FL, USA)

Department of Computer Science, Fall 2025 – Present

COP 3330: Object Oriented Programming — Fall 2025 (2 sections, ~150 students/section)

CAP 3103: Foundations of Human Computer Interaction — Spring 2026 (1 section, ~130 students)

COT 3100C: Discrete Structures I — Spring 2026 (1 section, ~160 students)

COT 4210: Discrete Structures II — Fall 2025 (1 section, 150 students), Spring 2026 (2 sections, ~330 students total)

Instructor of Record

University of North Florida (Jacksonville, FL, USA)

School of Computing, Fall 2023 – Spring 2025

COP 2220: Programming I

COP 3530: Data Structures

COT 3210: Theory of Computation

CIS 4930: Special Topics in Novel User Interfaces

Teaching Assistant

University of Central Florida (Orlando, FL, USA)

Department of Electrical Engineering and Computer Science, August 2012 – May 2020

COP 3223C: Introduction to Programming with C (Tanvir Ahmed) – Spring 2020

CAP 3104: Foundations of HCI (Michael McAlpin) – Fall 2019

COP 2500: Introduction to Computer Science (Jerry Hensel) – Spring 2013

COT 3100C: Introduction to Discrete Structures (Shaojie Zhang) – Fall 2012

Tasks: Taught lab sections, held office hours, proctored examinations, and graded projects, homework, and programming assignments across all courses.

Research Projects

Augmented Reality for Physics Education

University of Central Florida (Orlando, FL, USA)

January 2016 – May 2021

Low-cost augmented reality displays have enabled developers to target new domains for applications. The education domain, specifically physics, benefits from the interaction and integration enabled by AR interfaces. The primary goal of this project is to enable distance learning students to experience immersive labs which interact with their surroundings and provide sandbox tools for non-technical educators to develop their experiments for students. My contributions include project conception, simulation development (Unity3D/C#), and sandbox development.

Visualizing Uncertainty Data for Naval Planning

University of Central Florida (Orlando, FL, USA)

December 2015 – May 2019

Tactical planning requires a careful balance of minimizing risks and maximizing rewards. One way we augment decision-making processes is with visualizations that distill uncertainty related to gathered intelligence and situation reports. This project, which is funded by the Office of Naval Research, explores the cognitive effects of various levels of uncertainty on the decision-making process in several domains. My coding contributions include rapid prototyping web-based clients (HTML/JS/CSS), developing tactical simulation testbeds (Unity3D/C#), and managing experimental data (Python/R).

Alternative Controls for Multirotor UAVs

Microsoft Research (Redmond, WA, USA)

June 2015 – September 2015

Consumer UAVs have become more ubiquitous, but the control schemes behind them have not changed since their introduction. Currently, multirotor UAV borrow their control layouts from RC planes, which is not an intuitive mapping. The goal of this project was to define and implement simpler controls for the casual user, instead of focusing on enthusiasts, by utilizing a common game controller and skeletal tracking. My coding contributions included flight management code (C++), ground station app (C#), and wireless communication (C++ with ROS).

Improving \$-Family Recognizers

University of Central Florida (Orlando, FL, USA)

February 2015 – December 2020

Sketch recognition has focused heavily on template-based (kNN) recognition in recent years. Dollar family recognizers are typically leveraged for their balanced simplicity and accuracy. This project aims to determine the effect of dataset pruning and stroke averaging on the spatial and temporal performance of \$-family recognizers. My coding contributions included dataset management code (C#), stroke manipulation and processing (C#), and experimental management and analysis (Python).

Doppler Effect Gesture Recognition

University of Central Florida (Orlando, FL, USA)

February 2014 – December 2016

Current gesture recognition technology requires the use of electromagnetic trackers attached to the body or cameras placed in the environment. There are work environments where noise or security issues would prohibit the use of these devices. This project focused on using the Doppler Effect to detect coarse gestures using sounds emitted from speakers placed around a user. My coding contributions included the application layout (WPF), audio processing (C#), and gesture recognition logic (C#).

Head-Tracking for UAV Teleoperation

University of Central Florida (Orlando, FL, USA)

March 2013 – February 2014

Manipulating robots using gestural inputs has been studied often in recent years, with an emphasis on interactivity and naturalness. One mode of control that has not been studied as the primary navigation control is head tracking. This project focused on using head tracking to control an Unmanned Aerial Vehicle (UAV) in an indoor environment. Five control schemes were developed and compared to a ubiquitous game controller. My contribution to this work was developing ROS nodes for the head tracker (C++), HMD (C++/OpenGL), and control node (C++), as well as conducting the user study and analyzing the resulting data using SPSS. This work was published in IUI 2014.

Multimodal Interfaces for Robot Control

University of Central Florida (Orlando, FL, USA)

December 2012 – September 2013

Typical control stations for a group of robots include a large number of joysticks and assorted buttons for switching control modes and targets. This project aimed to reduce the cognitive load on operators by using a combination of body and hand gestures that better represented the robots' movements than a simple button press. The three selected robots were a humanoid robot, a flying UAV, and a scorpion-like robot. My contributions to this project included a literature review and developing the UAV control class in C++.

Image Localization using Panoramio

University of Central Florida (Orlando, FL, USA)

May 2011 – October 2011

Image localization relies heavily on the density of the source dataset. Prior work using a systematically gathered dataset from Google Maps StreetView showed good performance for localization. This project aimed to test the performance of the previously developed system on a less structured dataset collected from image-sharing webpages such as Flickr, Picasa, and Panoramio. I developed web crawlers for each of the repositories in Python that queried the servers via their respective APIs. Analysis of the data was done using MATLAB. This work was part of the Research Experience for Undergraduates in Computer Vision program.

Publications

Conference Publications

- Taranta, E., **Pittman, C.**, Oakley, J., Maslych, M., Maghoumi M., & LaViola Jr, J. J. (April 2020). Moving Toward an Ecologically Valid Data Collection Protocol for 2D Gestures in Video Games. Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems.
- **Pittman, C.** & LaViola Jr, J. J. (March 2020). PhyAR: Determining the Utility of Augmented Reality for Physics Education in the Classroom. Proceedings of the 2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR).
- Song, J., Newton, O., Fiore, S., **Pittman, C.**, LaViola Jr, J. J. (September 2019). Examining Training Comprehension and External Cognition in Evaluations of Uncertainty Visualizations to Support Decision Making. Proceedings of the Human Factors and Ergonomics Society Annual Meeting
- Taranta, E., Koh, S., Williamson, B., Pfeil, K., **Pittman, C.**, & LaViola Jr, J. J. (May 2019). Pitch Pipe: An Automatic Low-pass Filter Calibration Technique for Pointing Tasks. Proceedings of the 45th Graphics Interface Conference.
- **Pittman, C.** & LaViola Jr, J. J. (March 2019). Determining Design Requirements for AR Physics Education Applications. Proceedings of the 2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR).
- Song, J., Newton, O., Fiore, S., Coad, J., Clark, J., **Pittman, C.**, & LaViola Jr, J. J. (September 2018). Examining the Impact of Training and Feedback on Visualization-Supported Decision Making under Uncertainty. Proceedings of the Human Factors and Ergonomics Society Annual Meeting.
- Fiore, S., Song, J., Newton, O., **Pittman, C.**, Warta, S., & LaViola Jr, J. J. (July 2018). Determining the effect of training on uncertainty visualization evaluations. International Conference on Applied Human Factors and Ergonomics.

- **Pittman, C.** & LaViola Jr, J. J. (May 2017). Multiwave: Complex Hand Gesture Recognition Using the Doppler Effect. Proceedings of the 43rd Graphics Interface Conference.
- Taranta, E., Samiei A., Maghoumi, M., Khaloo, P., **Pittman, C.**, & LaViola Jr, J. J. (May 2017). Jackknife: A Reliable Recognizer with Few Samples and Many Modalities. Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems. **[Best Paper Honorable Mention]**
- Taranta, E., Maghoumi, M., **Pittman, C.**, & Laviola Jr, J. J. (October 2016). A Rapid Prototyping Approach to Synthetic Data Generation for Improved 2D Gesture Recognition. Proceedings of the 29th Annual Symposium on User Interface Software and Technology.
- **Pittman, C.**, Wisniewski, P, Brooks, C., & LaViola Jr, J. (May 2016). Multiwave: Doppler Effect Based Gesture Recognition in Multiple Dimensions, Proceedings of the 34th Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems.
- **Pittman, C.**, Taranta. E., and LaViola, J. (March 2016). A \$-Family Friendly Approach to Prototype Selection, Proceedings of the 2016 ACM International Conference on Intelligent User Interfaces.
- **Pittman, C.** and LaViola, J. (February 2014). Exploring Head Tracked Head Mounted Displays for First Person Robot Teleoperation, Proceedings of the 2014 ACM International Conference on Intelligent User Interfaces.

Journal Papers

- Taranta, E., **Pittman, C.**, Maghoumi M., Maslych, M., Moolenaar, Y., & LaViola Jr, J. J. (January 2021). Machete: Easy, Efficient, and Precise Continuous Custom Gesture Segmentation. ACM Transactions on Computer-Human Interaction (TOCHI) 28.1 (2021): 1-46.

Book Chapters

- LaViola Jr, J. J., Buchanan, S., & **Pittman, C.** (2014). Multimodal Input for Perceptual User Interfaces. Interactive Displays: Natural Human-Interface Technologies.

Patents

- Taranta, E., Maghoumi, M., Pittman, C., & Laviola Jr, J. J. (2018), Synthetic Data Generation of Time Series Data, US Patent App. 15/651,219

Students Advised

Undergraduate Research Assistants

- Ertis Seferi - University of North Florida, 2024–2025
- John Reddick - University of North Florida, 2024–2025
- Stephen Pond - University of North Florida, 2024–2025

Master's Thesis Committee Member

- Conor Nolan (Chair) - University of North Florida, 2024–2025
- Ahmed Sayed (Member) - University of North Florida, 2024–2025
- Nevzat Demirseren (Co-chair) - University of North Florida, 2024–2025

Professional Activities

University & Departmental Service

- Faculty Search Committee Member, University of Central Florida, Fall 2025 – Present
- Faculty Search Committee Member, University of North Florida, Fall 2024 – Spring 2025

Professional Development

- Writing Across the Curriculum (WAC) AI Fellows, University of Central Florida, Spring 2026 Participant
- Knighted Faculty Program, University of Central Florida, Spring 2026 Participant

Paper Reviewer

- Program Committee member and Reviewer (9 papers), the 24th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2025)
- Conference Full Paper Reviewer (1 paper), the 2025 ACM Symposium on Spatial User Interaction (SUI 2025)
- Program Committee member and Reviewer (8 papers), the 23rd IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2024)
- Journal Paper Reviewer (1 paper), Software: Practice and Experience 2021.
- Conference Full Paper Reviewer (1 paper), the 2020 CHI Conference on Human Factors in Computing Systems (CHI 2020)
- Conference Full Paper Reviewer (1 paper), the 18th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2019)
- Conference Full Paper Reviewer (2 papers), the 17th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2018)
- Journal Paper Reviewer (1 paper), Virtual Reality (Springer 2018)
- Conference Full Paper Reviewer (1 paper), the 2017 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY 2017)
- Conference Poster Reviewer (3 posters), the 14th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2015)
- Conference Full Paper Reviewer (1 paper), the 13th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2014)

Presentations

- Poster presenter for "PhyAR: Determining the Utility of Augmented Reality for Physics Education in the Classroom" at the 2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), Remote, May 2020
- Presenter for "Pitch Pipe: An Automatic Low-pass Filter Calibration Technique for Pointing Tasks" at Graphics Interface Conference (GI), Kingston, ON, Canada, May 2019
- Presenter for "Multiwave: Complex Hand Gesture Recognition Using the Doppler Effect" at Graphics Interface Conference (GI), Edmonton, AB, Canada, May 2017
- Poster presenter for "Multiwave: Doppler Effect Based Gesture Recognition in Multiple Dimensions" at Annual ACM on Human Factors in Computing Systems (CHI), San Jose, CA, May 2016
- Presenter for "A \$-Family Friendly Approach to Prototype Selection" at International Conference on Intelligent User Interfaces (IUI), Sonoma, CA, Mar. 2016
- Poster presenter for "Exploring Head Tracked Head Mounted Displays for First Person Robot Teleoperation" at International Conference on Intelligent User Interfaces (IUI), Haifa, Israel, Feb. 2014

Conferences Attended

- Sep 2024: 2024 Florida Digital Humanities Consortium (FLDH) Annual Conference, Orlando, FL
- May 2020: IEEE Virtual Reality 2020 (IEEE-VR), Atlanta, FL (Remote)
- May 2019: Graphics Interface Conference (GI), Kingston, ON, Canada
- April 2018: Conference on Human Factors in Computer Systems (CHI), Montreal, QC, Canada
- May 2017: Graphics Interface Conference (GI), Edmonton, AB, Canada
- May 2017: Conference on Human Factors in Computer Systems (CHI), Denver, CO

- May 2016: Conference on Human Factors in Computer Systems (CHI), San Jose, CA
- Mar 2016: International Conference on Intelligent User Interfaces (IUI), Sonoma, CA
- Apr 2014: Conference on Human Factors in Computer Systems (CHI), Toronto, ON, Canada
- Feb 2014: International Conference on Intelligent User Interfaces (IUI), Haifa, Israel
- Apr 2013: IEEE Virtual Reality 2013 (IEEE-VR), Orlando, FL

Awards

- 2020 McKnight Doctoral Fellowship recipient

Skills

Programming Languages

C/C++, C#, Java, Python

Web Programming

HTML, JavaScript, CSS, OpenLayers3

Libraries & Tools

Unity3D, Adobe Creative Suite, MATLAB, SPSS, R, ROS, V-Rep, OpenGL, CUDA, Kinect SDK

Relevant Coursework

- **Doctorate:** Computer Graphics (WebGL), Bioinformatics, Computer Vision, Computational Complexity
- **Master's:** Algorithms & Analysis, Pen-Based User Interfaces, 3D User Interfaces, 3D Computer Vision, Evolutionary Computation, Parallel Programming
- **Bachelor's:** Robot Vision, Computer Graphics (OpenGL), AI for Game Programming, Linear Algebra, Differential Equations